



Introduction to Pediatric Cardiac Physical Examination

DR. LAILA TUTUNJI

Framework

- ▶ The Physical Examination follows a general framework
- ▶ Regardless of the system being examined, It always starts with
 1. General Inspection
 2. Vital Signs
 3. Weight and Height
 4. Exposure
 5. Inspection
 6. Palpation
 7. Percussion
 8. Auscultation
 9. Special Maneuvers
 10. Related Systems or Organs

Pediatric Ages and Stages

- ▶ Based on Growth and Development the pediatric age is divided into stages:
- ▶ What is growth?
- ▶ What is development?
- ▶ Pediatric patients are NOT small adults
- ▶ Every Stage has its own normal values and approach

Pediatric Ages and Stages

- ▶ Pediatric Stages:
- ▶ Neonate (first 28 days of life)
- ▶ Infant (1 month to 1 year)
- ▶ Toddler (1-3 years)
- ▶ Early Childhood (3-5 years)
- ▶ Childhood (6-11 years)
- ▶ Adolescence (12-21 years) divided into early 12-15 years, middle 16-18 years and late adolescence 18-21 years.

Physical Examination : General Inspection

- ▶ Level of Consciousness
- ▶ Signs of Respiratory Distress:
 1. Dyspnea
 2. Tachypnea
 3. Use of Accessory Muscles
 4. Grunting
 5. Cyanosis
- ▶ Colors: Jaundice, Pallor, Malor Flush
- ▶ Dysmorphic Features

Physical Examination : Vital Signs

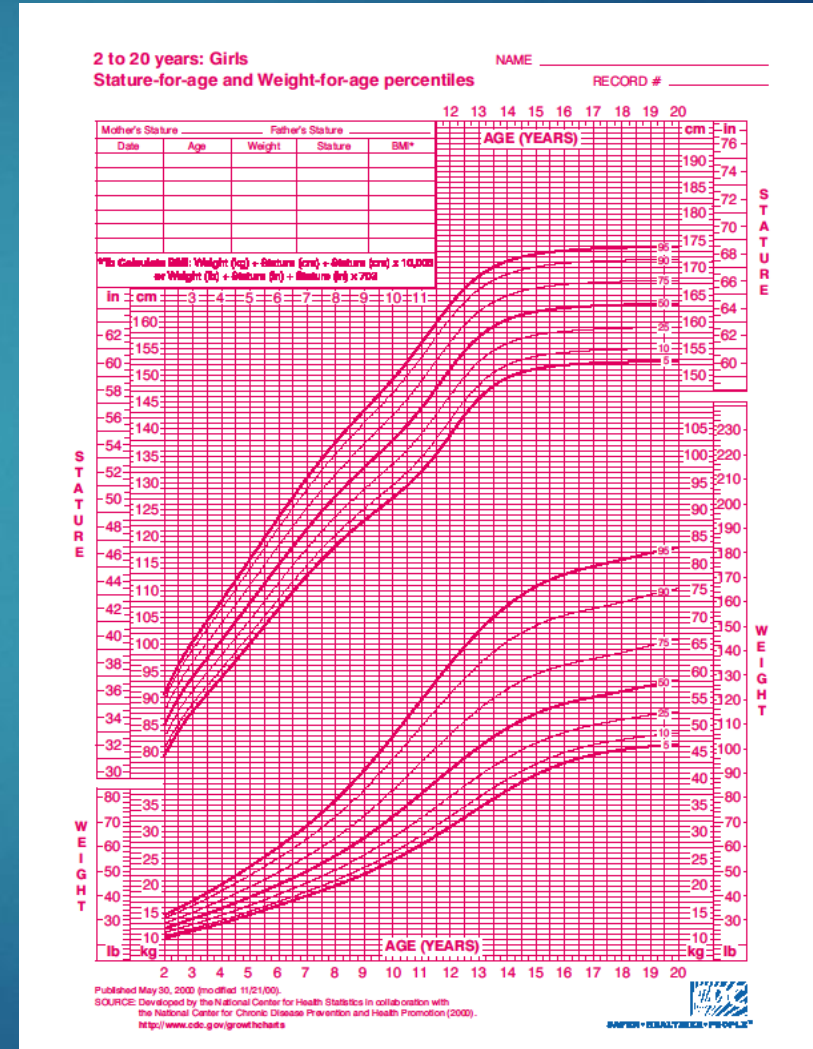
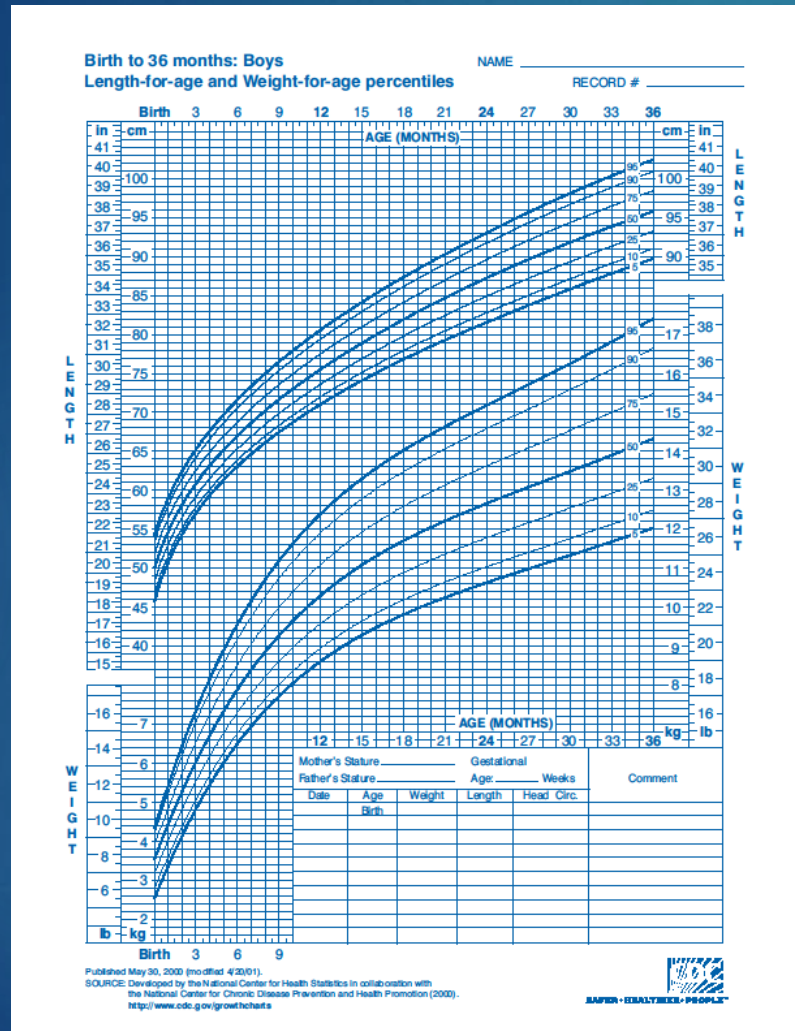
- ▶ Temperature
- ▶ Heart Rate
- ▶ Respiratory Rate
- ▶ Blood Pressure

- ▶ Additional:
- ▶ Pulse Oximetry
- ▶ Pain Score

Physical Examination : Weight and Height

- ▶ Growth Curves
- ▶ Normality defined by percentiles
- ▶ Normal growth is a weight and height between the 5% and 90% percentiles
- ▶ Major percentile lines 5%, 10%, 25%, 50%, 75%, 90% and 95%
- ▶ Less than the 5% is failure to thrive and more than 95% needs investigation
- ▶ A drop of two major percentile lines is considered failure to thrive

Sample Growth Curves



Physical Examination: Exposure of CVS System

► **Inspection:**

► From Foot of Bed

1. Signs of Respiratory Distress:
2. Intercostal and Sub costal retractions
3. Symmetry
4. Deformity
5. Skin
6. Scars

► **Palpation**

► Valvar Area Palpation :

- With the palm,
- looking for taps, or thrills

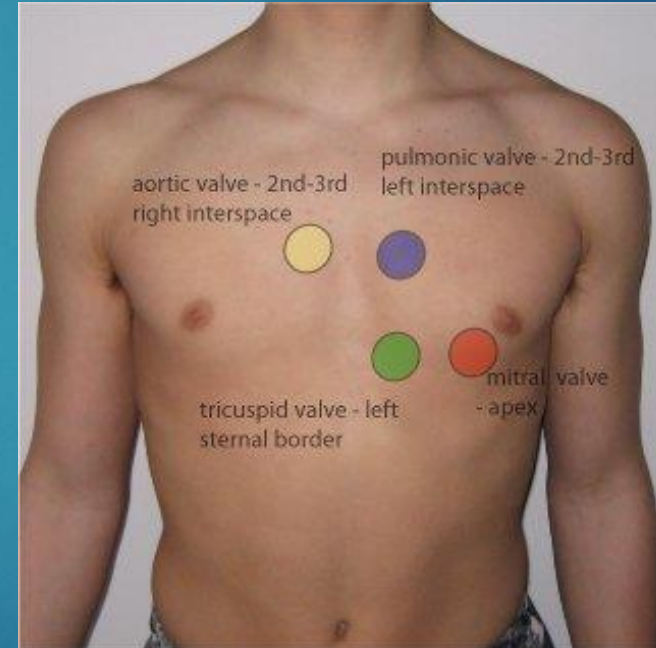
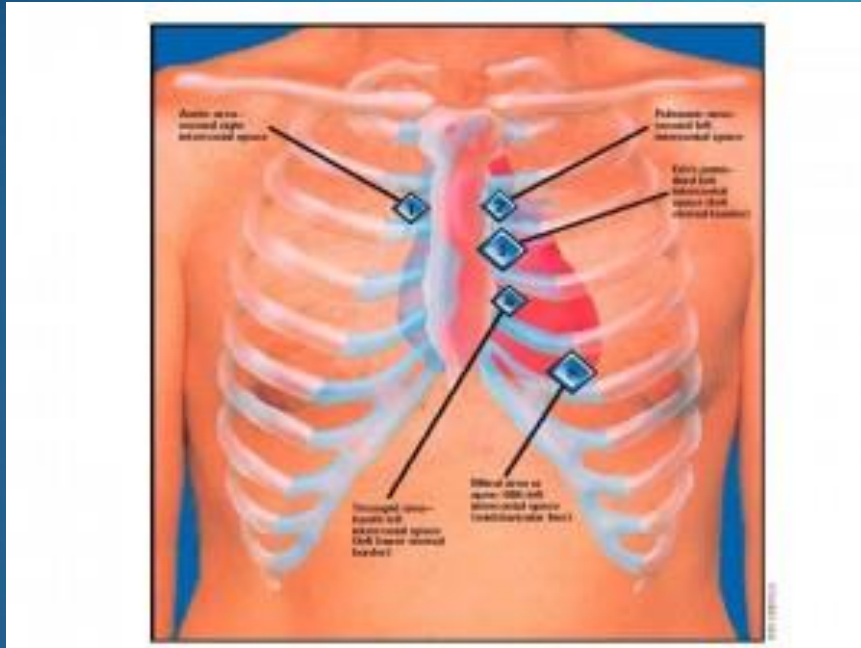
► Palpation of the Ventricles:

- With the whole hand
- Looking for RV heave
- LV heave

► Palpation of the Apex Beat

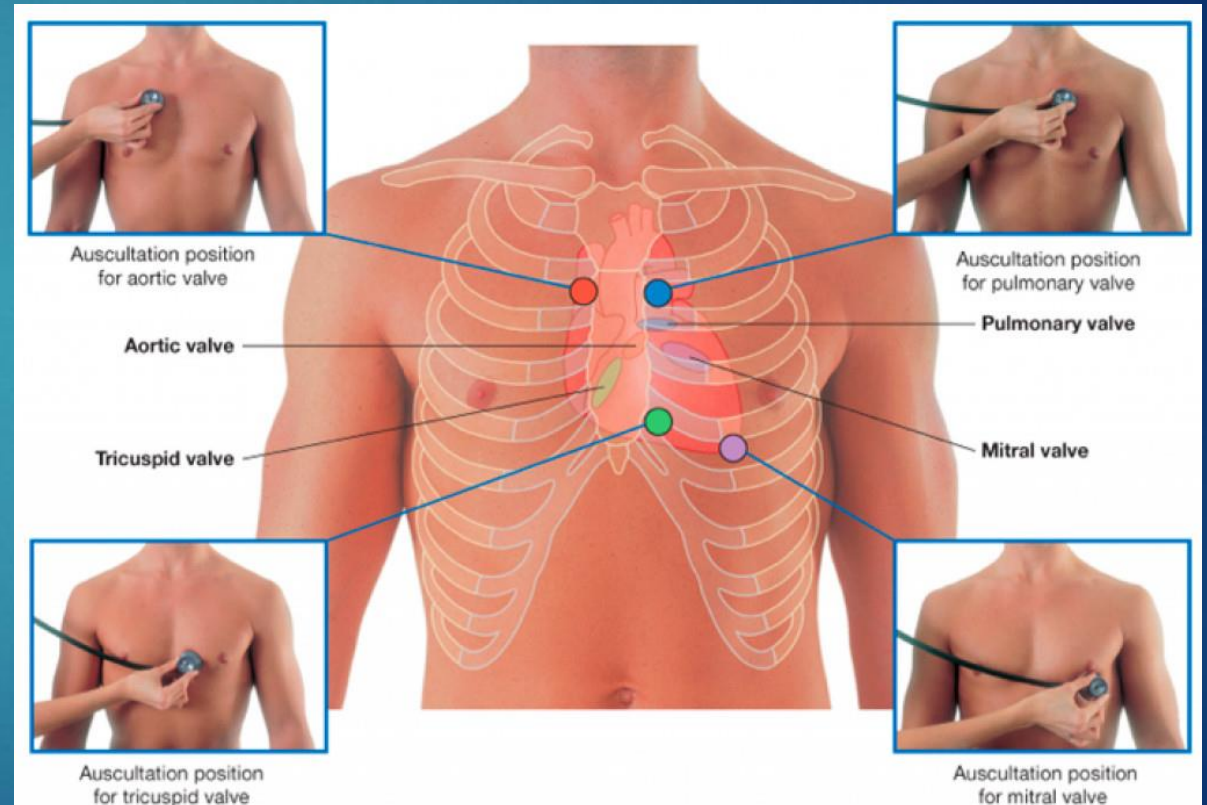
- With the tips of the fingers
- Looking for location and a gently tapping beat

CVS Palpation: Valves, Ventricles and Apex Beat



CVS Examination: Percussion and Auscultation

- ▶ **Percussion**
- ▶ Skipped in CVS physical Examination
- ▶ **Auscultation:**
- ▶ Valvar areas
- ▶ Bell and diaphragm
- ▶ Supine and Sitting
- ▶ Sites of radiation: neck and back and axilla
- ▶ S1 S2
- ▶ Murmurs
- ▶ Added Sounds



CVS Examination : Special Maneuvers

- ▶ Right sided structures are anterior and so need no maneuvers
- ▶ Left sided structures need maneuvers : Aortic Valve and Mitral Valve
- ▶ **Mitral Valve Maneuver:**
 - ▶ Left lateral decubitus position
 - ▶ Listen at the apex of the heart and the axilla
 - ▶ Systolic Regurgitant Murmur = MR
 - ▶ Early or mid Diastolic murmur = MS
 - ▶ Radiates to the Axilla
- ▶ **Aortic Valve Maneuver**
 - ▶ Leaning forward
 - ▶ Hold breath in expiration
 - ▶ Listen under the pulmonic valve area and the neck
 - ▶ Looking for the Aortic Regurgitation Murmur
 - ▶ Early Diastolic Decrescendo Murmur at the Left 3rd intercostal space and radiates to the neck

CVS Examination: Related Organs

- ▶ **The lung**

- ▶ Auscultation of bilateral lung fields
- ▶ Basal crackles or less frequently basal wheeze

- ▶ **The liver**

- ▶ Palpation
- ▶ +_ Liver Span

- ▶ **Signs of Heart Failure:**

- ▶ Tachycardia
- ▶ +_ tachypnea
- ▶ Hepatomegaly
- ▶ S3 or S4
- ▶ ? Gallop

CVS Examination: Vascular

- ▶ Palpation of two upper limb pulses
- ▶ Looking for volume, regularity, and delay
- ▶ Palpation of an upper limb and a lower limb pulse (usually right upper and right lower)
- ▶ Looking for volume, regularity and delay
- ▶ **Palpation for Lower Limb Edema**
- ▶ Shin of Tibea
- ▶ **Palpation of the capillary refill**
- ▶ Toe